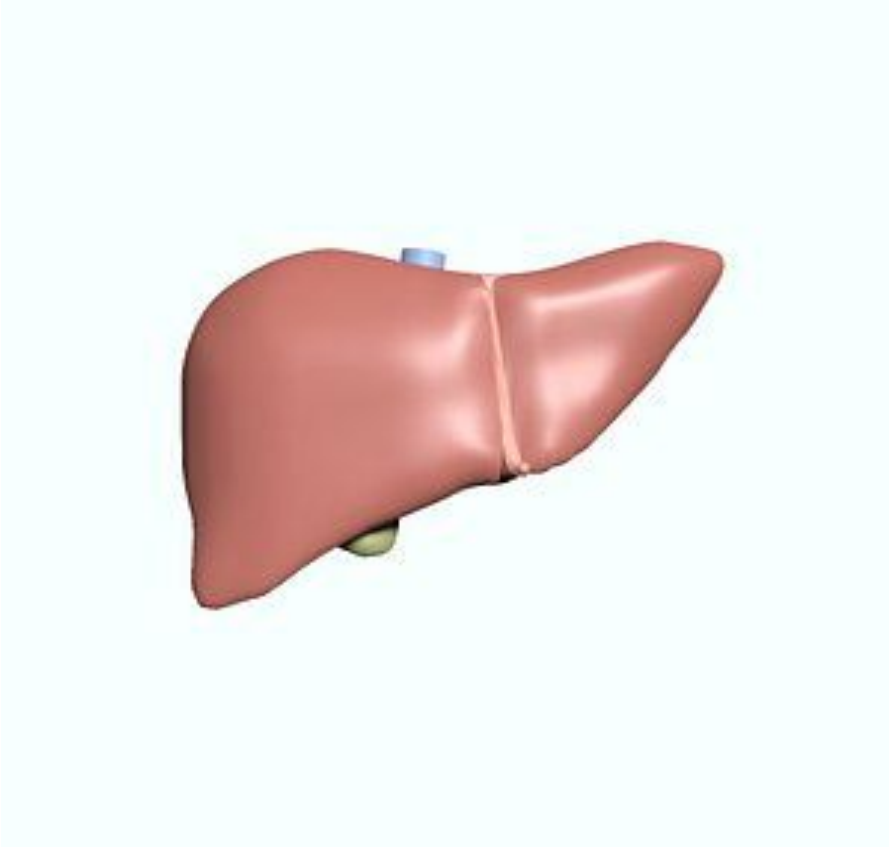
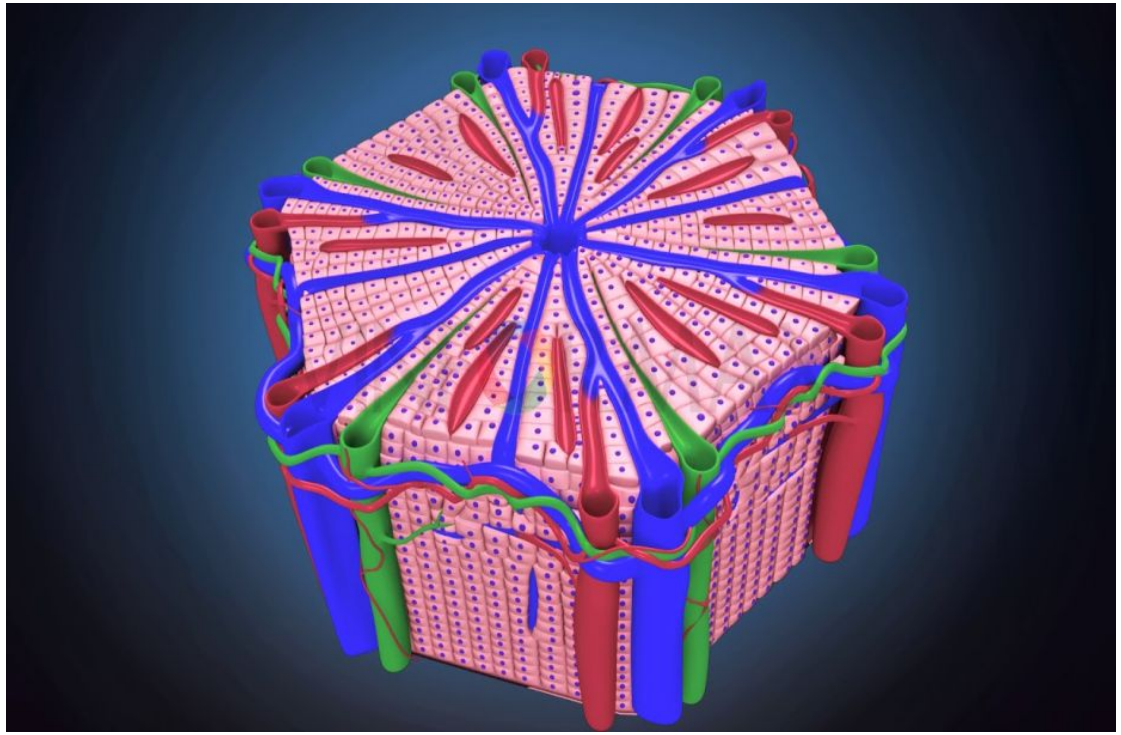
Histology of LIVER

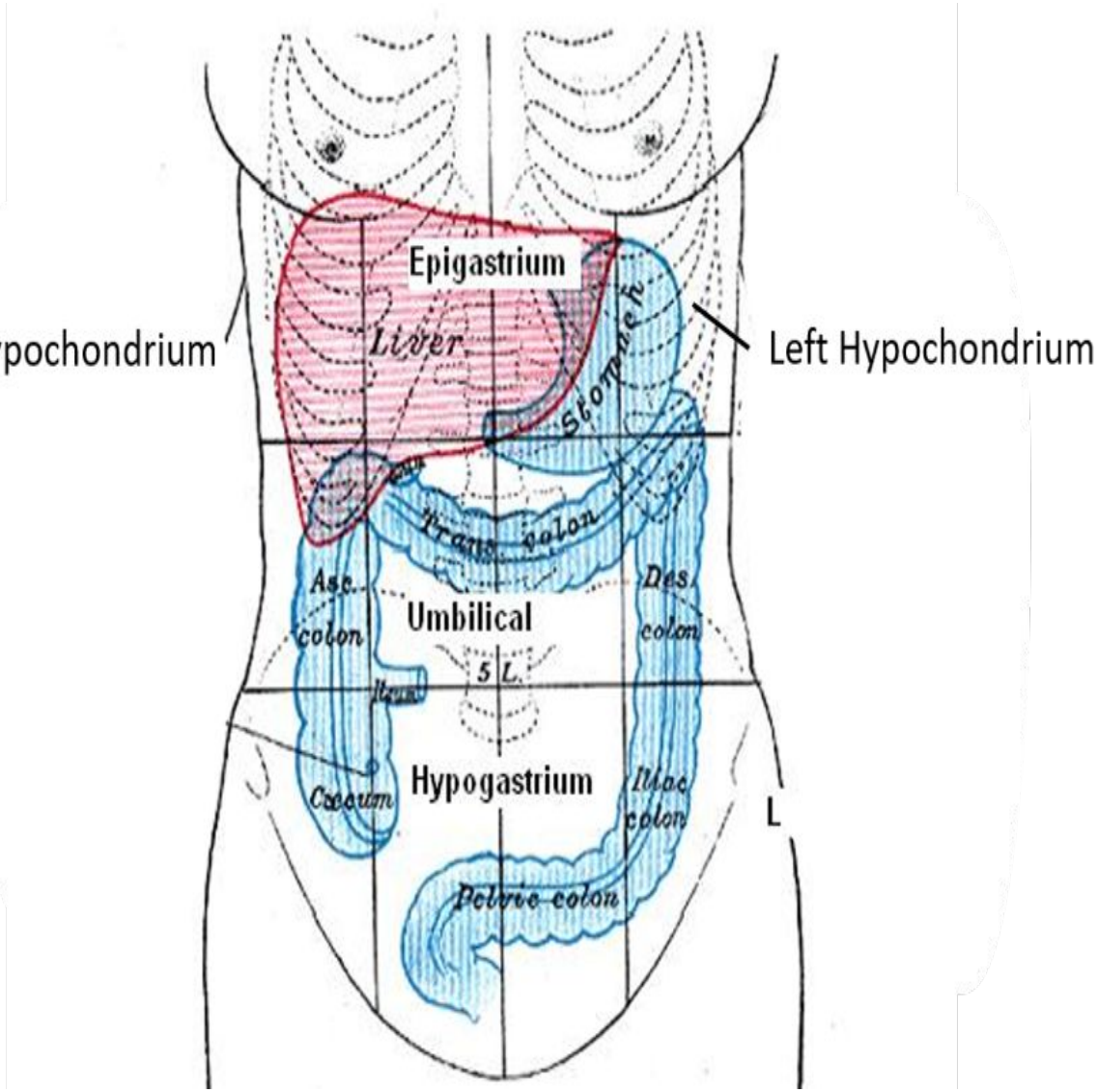




The Liver

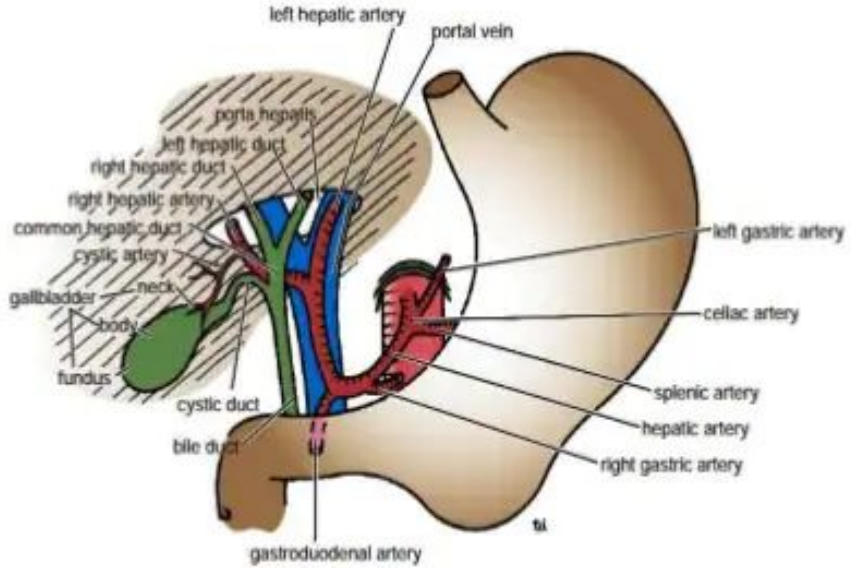
- Largest gland of the body.
- 1500 grams and 2.5% of total body weight.
- Location:

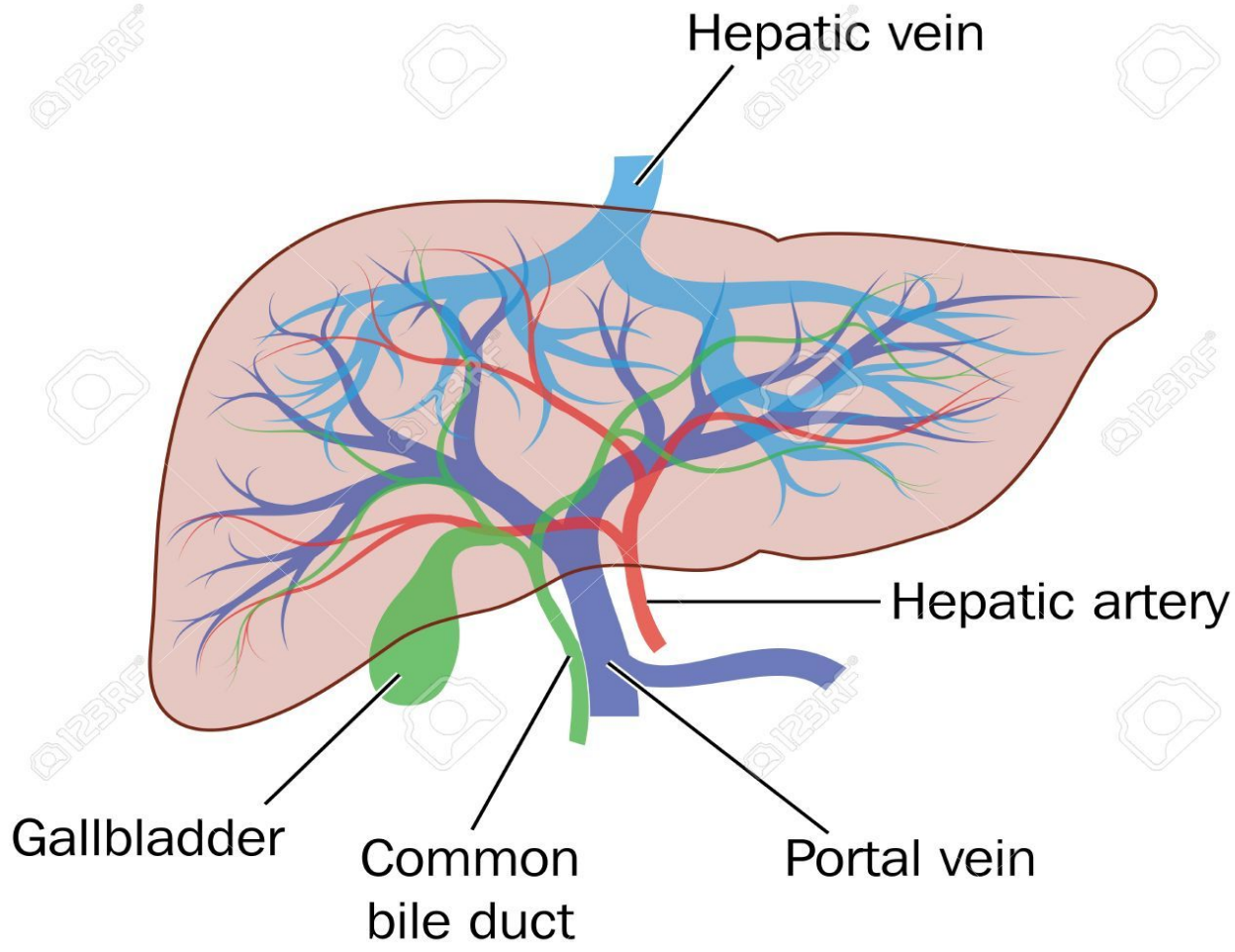
- Right hypochondrium
- Epigastric region
- Left hypochondrium



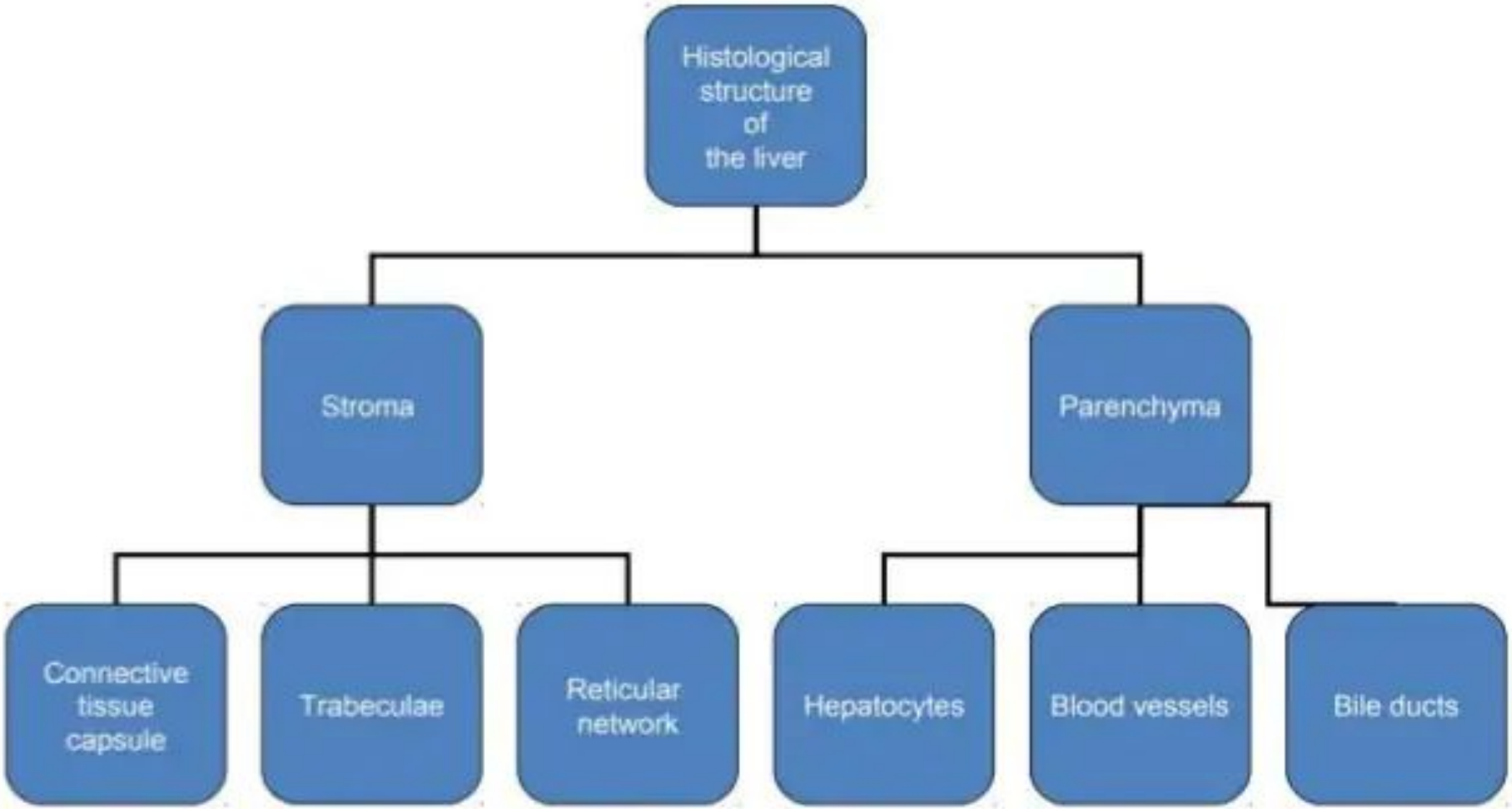
Vascular Supply of the Liver

- Receives dual vascular supply:
  - -Hepatic Portal Vein (75%)
  - - Hepatic Artery(25%)
- Both vessels enter the liver via Porta hepatis.





Histological Structure of the Liver

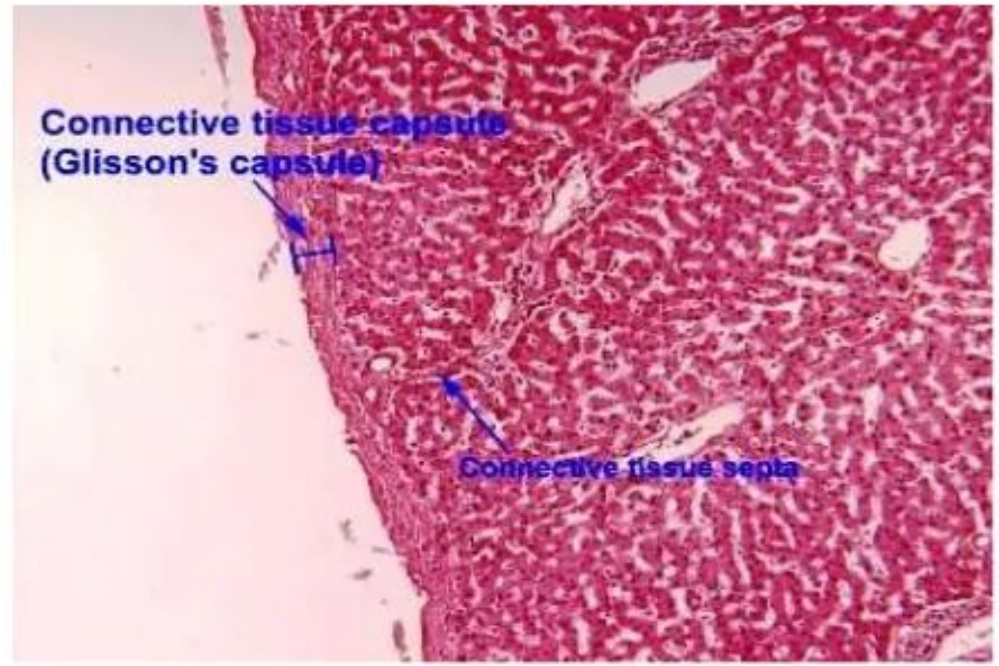


The Liver Stroma

- Consists of: - Connective Tissue Capsule
  - Trabeculae
  - Reticular fibers

1. Connective Tissue Capsule:

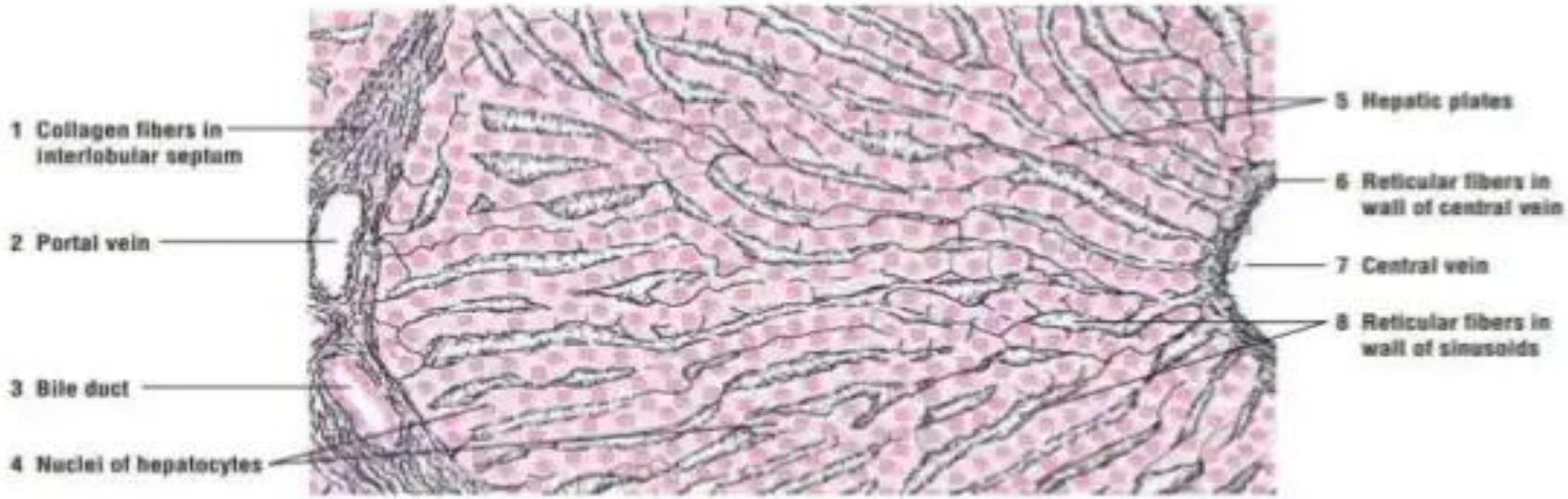
- Serous layer: Visceral Peritoneum
- Fibrous layer: Glisson’s capsule



- Trabeculae: The Glisson’s capsule extends into the interior of liver as numerous branching trabeculae and septa.

- Reticular fibers:

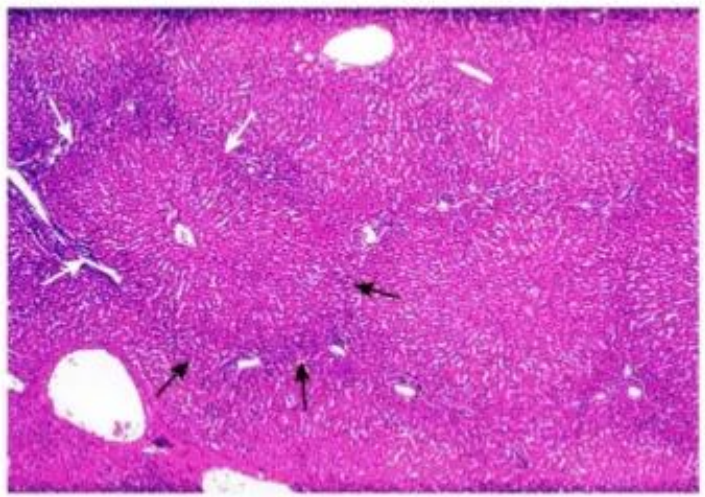
- Supporting connective tissue of the liver.
- Line the sinusoids, support the endothelial cells, and form a denser network of reticular fibers in the wall of the central vein.
- Also merge with the collagen fibers in the interlobular septum, where they surround the portal vein and the bile duct.

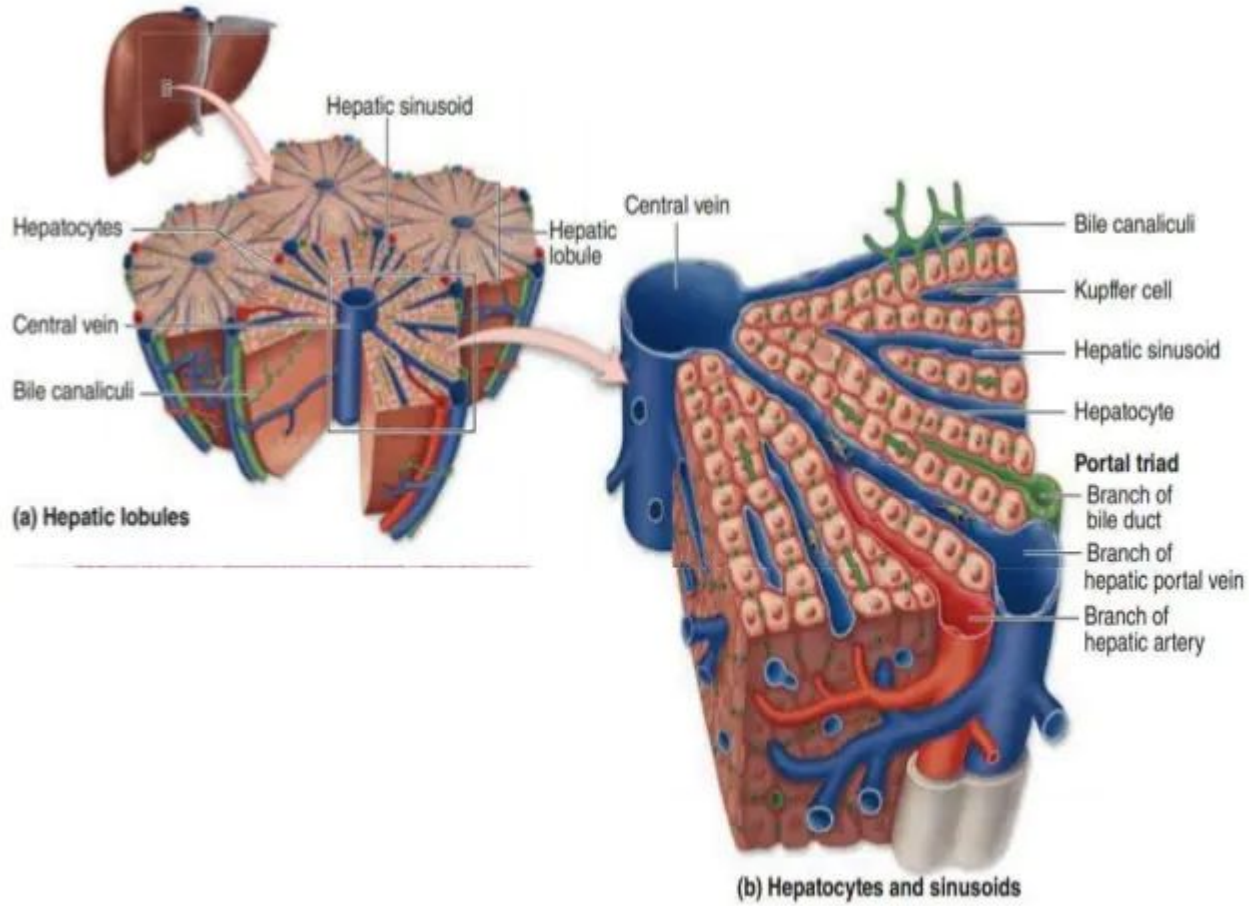


The Liver Parenchyma

- Organized as thousands of small hepatic lobules.

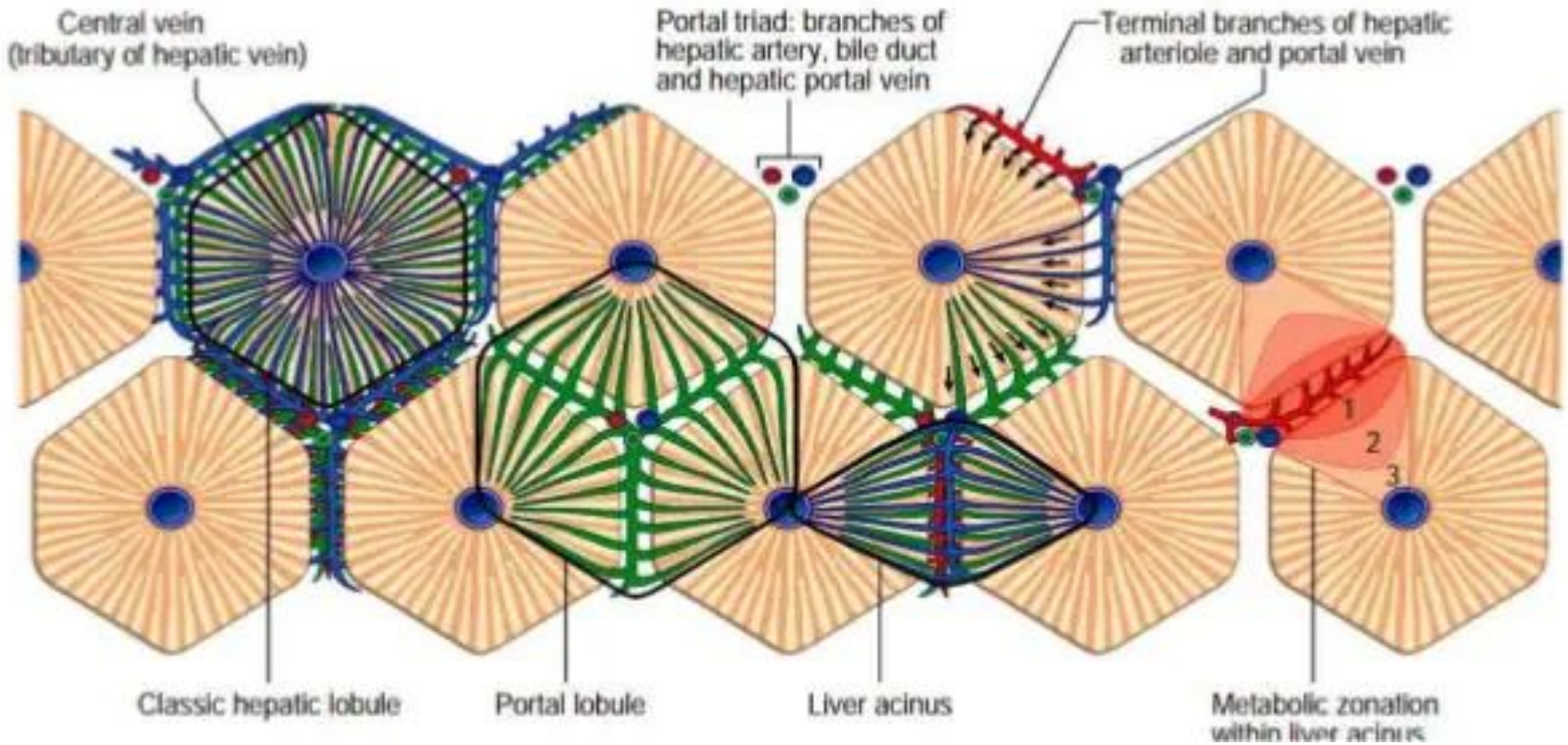
- **Hepatic Lobules**: Structural units of Liver.
  Roughly hexagonal arrangement of irregular plates or cords of hepatocytes radiating outward from a central vein.





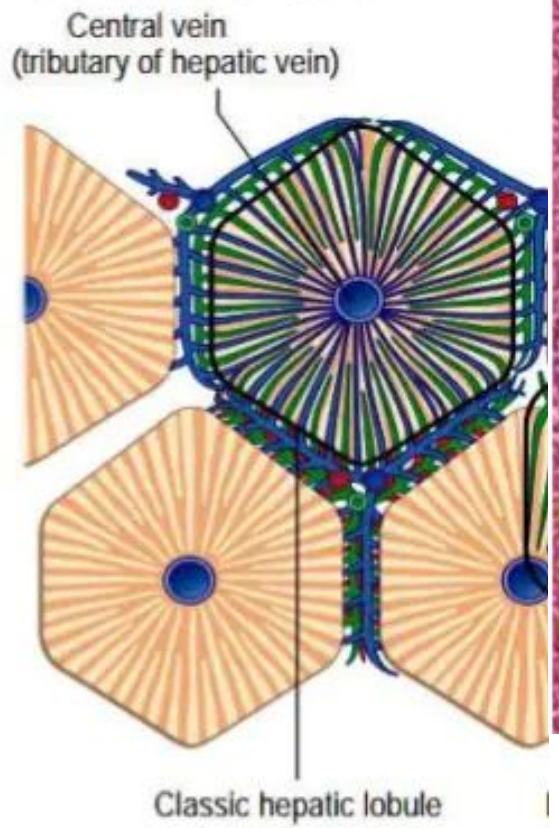
Concepts of Liver Lobules

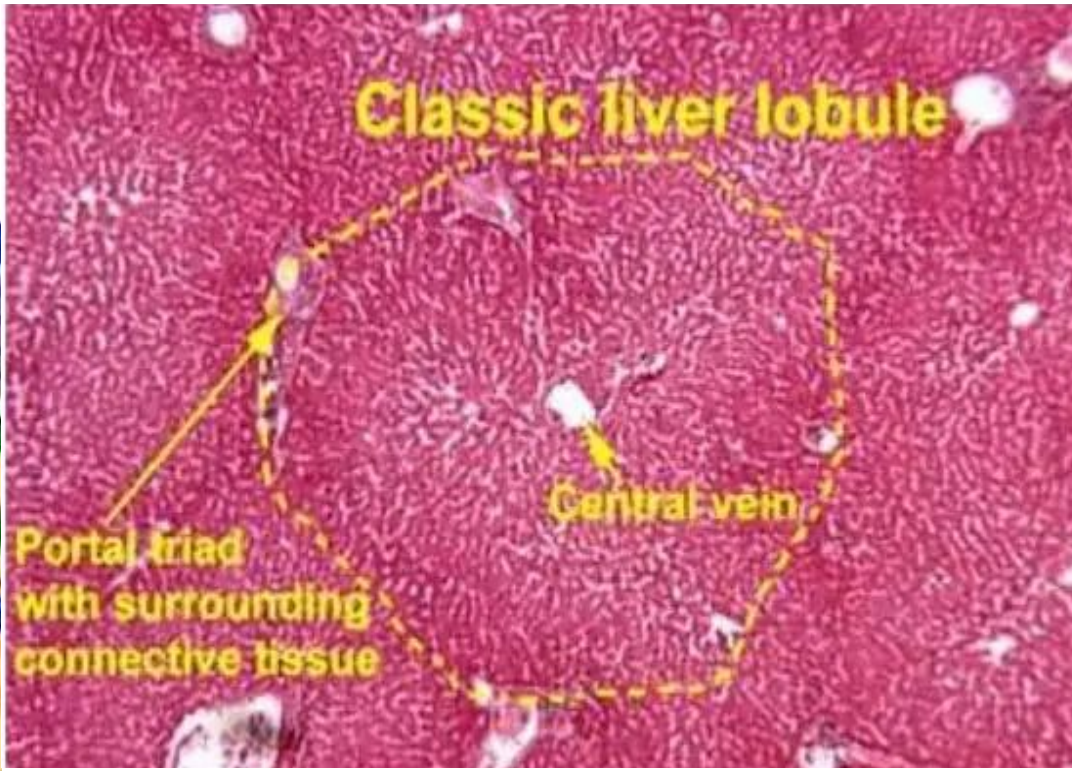
- Classical Hepatic Lobule
- Portal Lobule
- Hepatic Acinus of Rappaport



Classical Hepatic Lobule

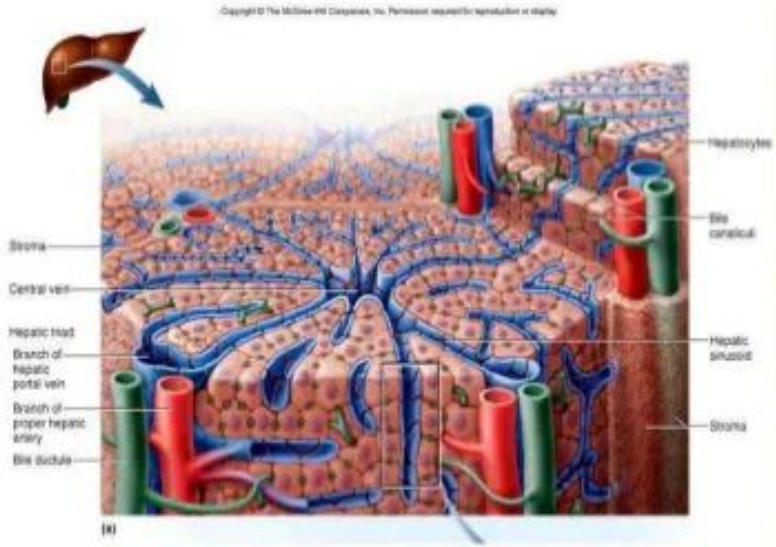
- Each lobule consists of a hexagonal mass of liver cells.
- Central axis occupied by: Central vein.
- Is an *independent venous unit*.
- From the central vein, hepatocytes radiate irregularly as plates known as Hepatic lamina.
- Spaces between the hepatic lamina are called Hepatic lacunae, occupied by Hepatic sinusoids.

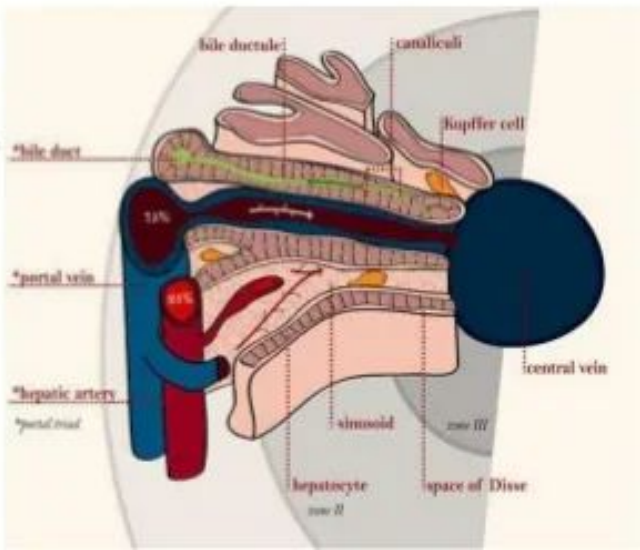




Classical hepatic lobule contd..

- Hepatic Sinusoids:
- Wide diameter capillaries.
- Their walls are fenestrated and made up of flattened endothelial cells.
- Kupffer cells and pit cells are attached to the endothelial surface.

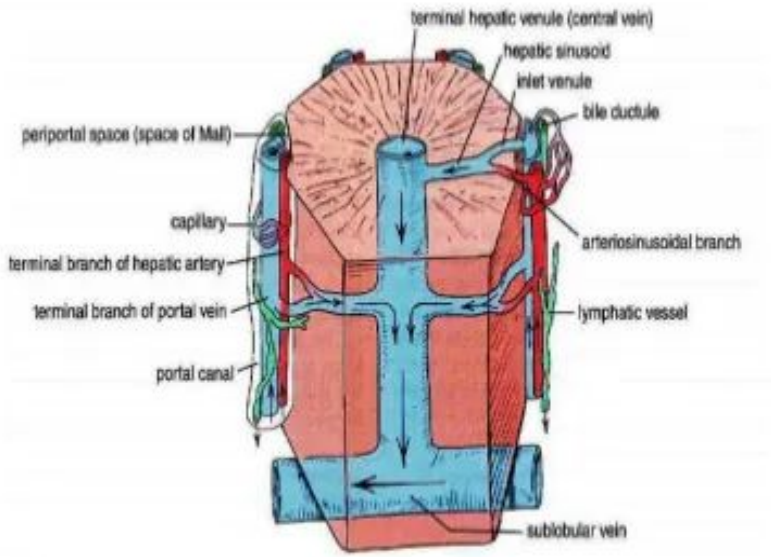






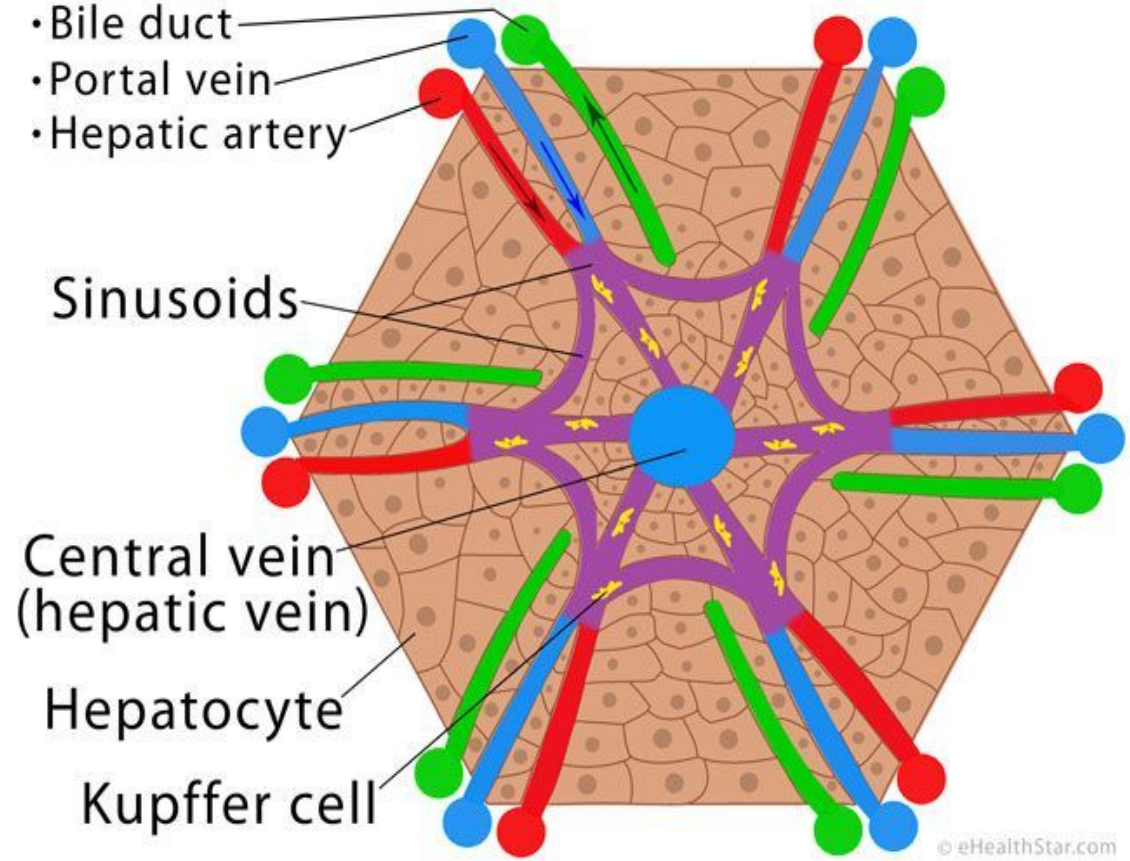
Classical hepatic lobule contd..

- Hepatic sinusoids receive a mixture of blood from the portal vein and the hepatic artery of adjacent portal area.
- They are interlaminar and centripetal in direction.
- The blood flows towards Central vein ⟶ Sublobular vein⟶Hepatic vein ⟶ Inferior Venacava.



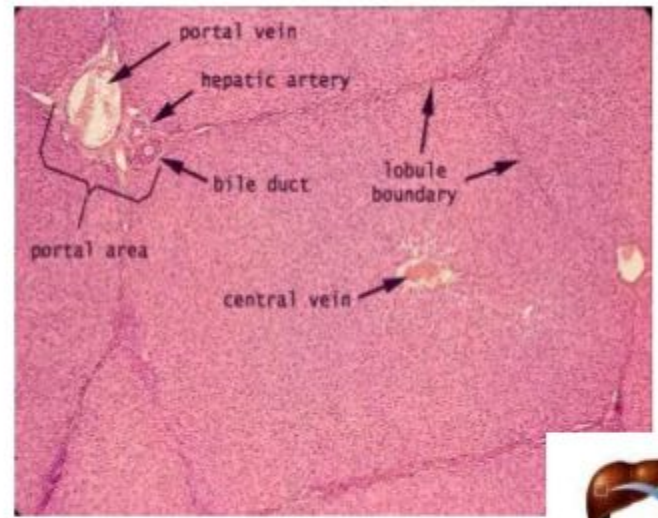
Hepatic Lobule

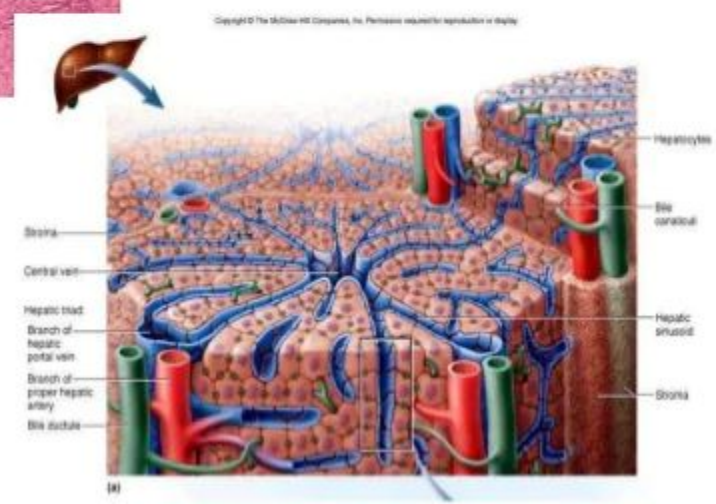
Portal triad:
- Bile duct
- Portal vein
- Hepatic artery



The Portal Area

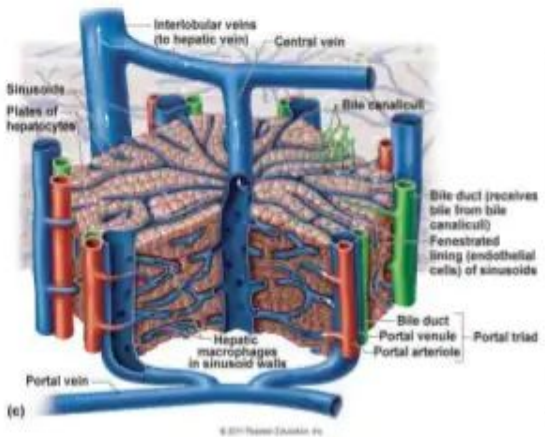
- Peripherally, each lobule has 3 to 6 portal areas with more fibrous connective tissue, each of which contains interlobular structures that comprise the portal triad. They include:
- A venule branch of the portal vein, with blood rich in nutrients but low in O2.
- An arteriole branch of the hepatic artery that supplies O2.
- One or two small bile ductules of cuboidal epithelium, branches of the bile conducting system.

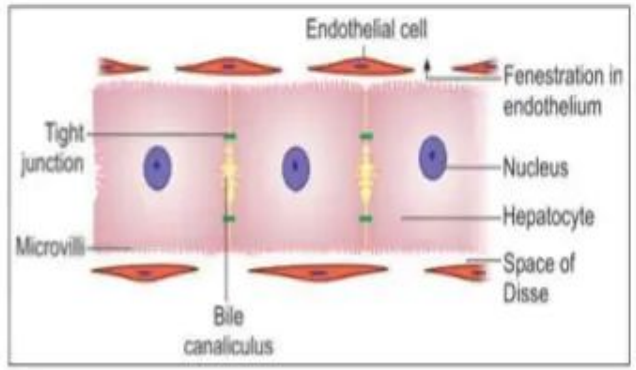




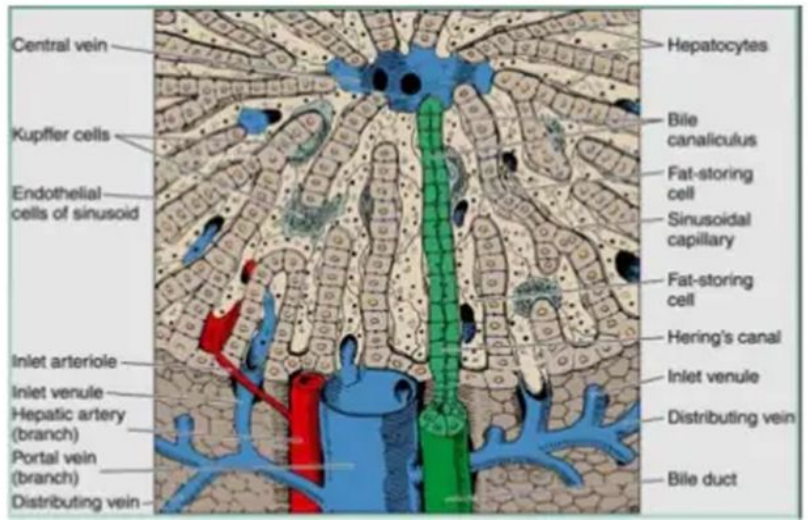
Bile Canaliculi

- Formed by spaces present between plasma membranes of adjacent liver cells.
- Form hexagonal networks around the liver cells.
- Borders around the canaliculi are sealed by tight junctions. This forms the blood-bile barrier.



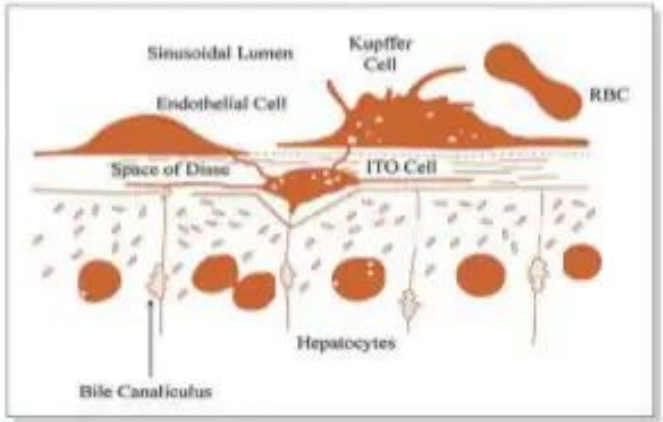


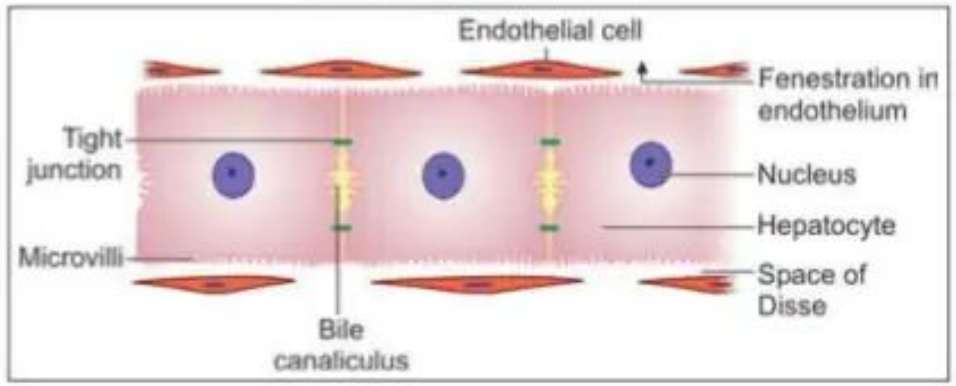
- The canaliculi pass to periphery of the hepatic lobules where they form intralobular canal of Herring, that finally drains into the interlobular duct of the portal area.
- Bile canaliculi are intralaminar and centrifugal in direction

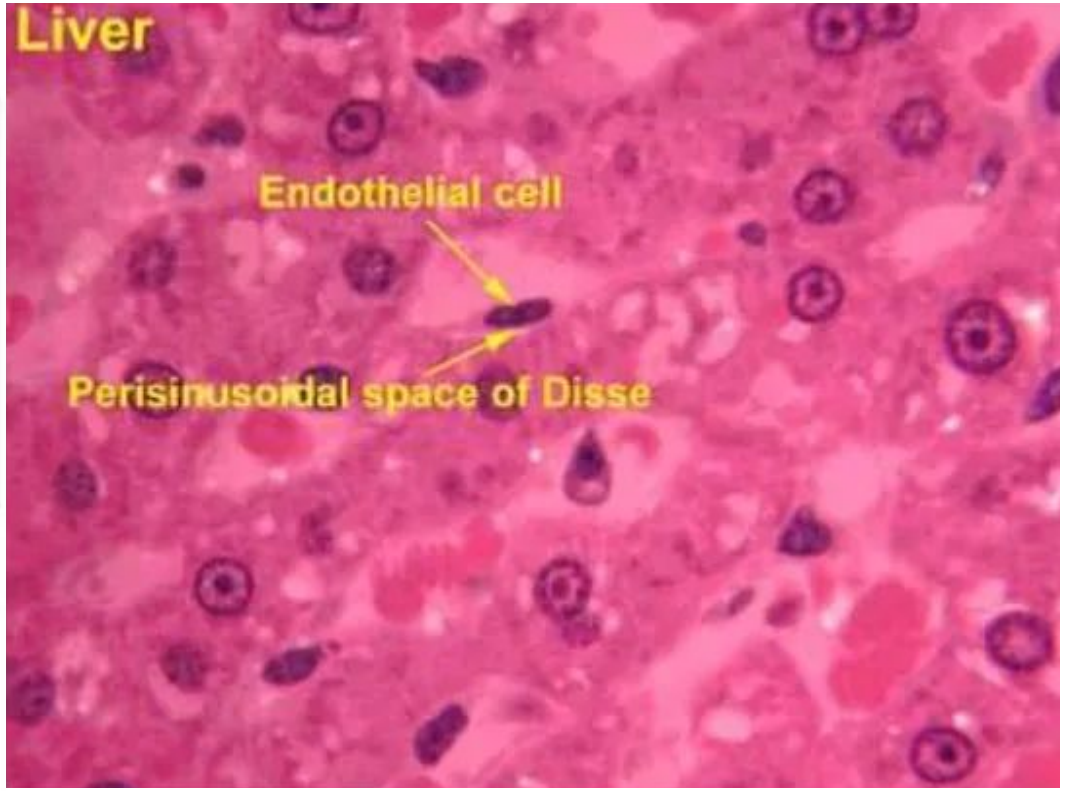


Space of Disse / Perisinusoidal Space

- Potential space between the wall of sinusoids and laminae of the liver cells.
- Filled with blood plasma and chylomicrons that percolate through the wall of sinusoids.
- Presence of *Ito cells*.







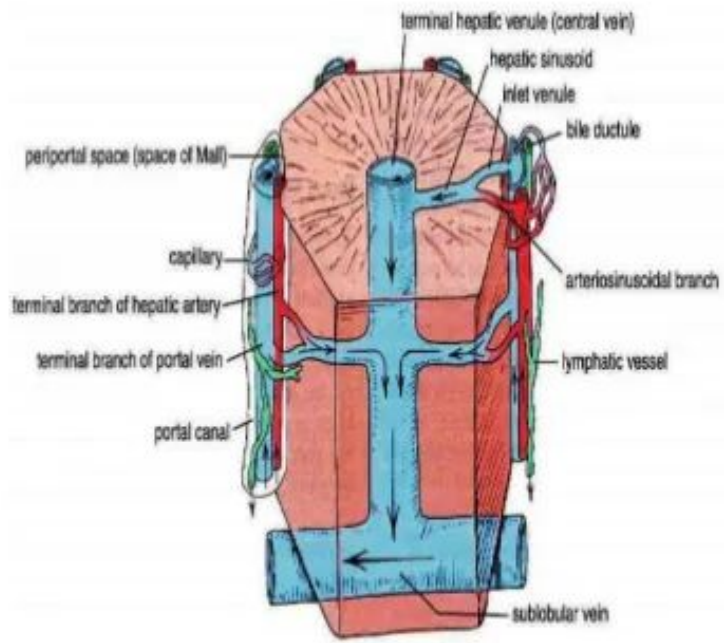
Ito cells

- Irregular outline with numerous lipid vesicle
- Function of Ito cells:

- Secrete collagenous matrix
- Provide growth factor for regeneration of damaged liver cells.
- Store Vitamin A in their lipid vesicles.

Space of Mall

- Potential space, between the glisson's capsule of portal area and the hepatic plates of the cells.
- Lymphatics of liver begin here.



Thank You

Pitch Deck Presentation October 2027